QC STORY PROJECT

Project Title:

To eliminate the Defect of Cargo bolt not fit for SC/DC Model.

Mahindra and Mahindra Ltd,NASIK_ Plant, _SCO/XYLO TCF.

	Plan	Actual
Start Date	1.05.2019	1.6.2019
Completion Date	5.11.19	5.11.19

Current Status: **Completed**

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summaries & Plan future actions
- Guidelines to user

Team information

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Nilesh Deore	Assistance Manager	Production	8888821705	Deore.nilesh@Mahindra.com
2	**Mentor**	Taj Shaikh	Sr Manager	Production	9850980299	Shaikh.taj@mahindra.com
3	**Member 1**	Sachin Bhamare	Deputy Manager	MQA	9158960111	Bhamabre.sachin@Mahindra.com
4	**Member 2**	Santosh Jadhav	Manager	SQA	9423303400	Jadhav.santosh@Mahindra.com
5	**Member 3**	Dilip Junnare	Senior Associate	Production	9923103633	JUNARE.DILIP@MAHINDRA.COM
6	**Member 4**	Madhav Hrdikar	Manager	CDMM	9922428949	Madhav.rdikar@Mahindra.com

Step1. Select Subject

Brief Description : **To eliminate the Defect of Cargo bolt not fit for SC/DC Model.**

What : Cargo bolt not fitted.

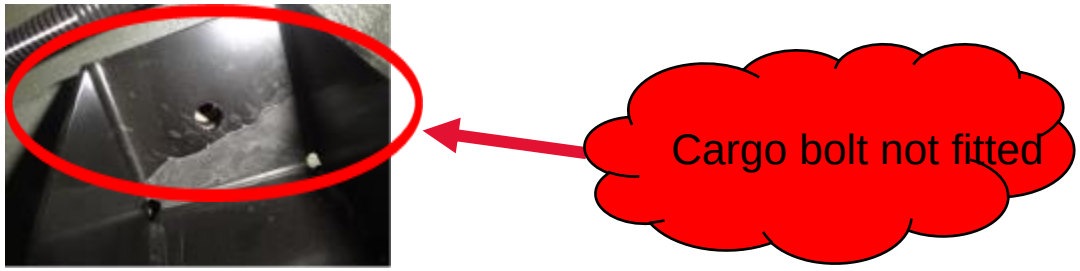
Where : At Cargo Drop Underpit stage at Final assembly line .

When : From Jun 19.

Who: By bolt tightening by associate at Final line assembly.

Why: Chronic Repeat Concern, Production loss, Quality loss, Cost loss, Deliver loss.

How: Avg. 14 cases / Month.



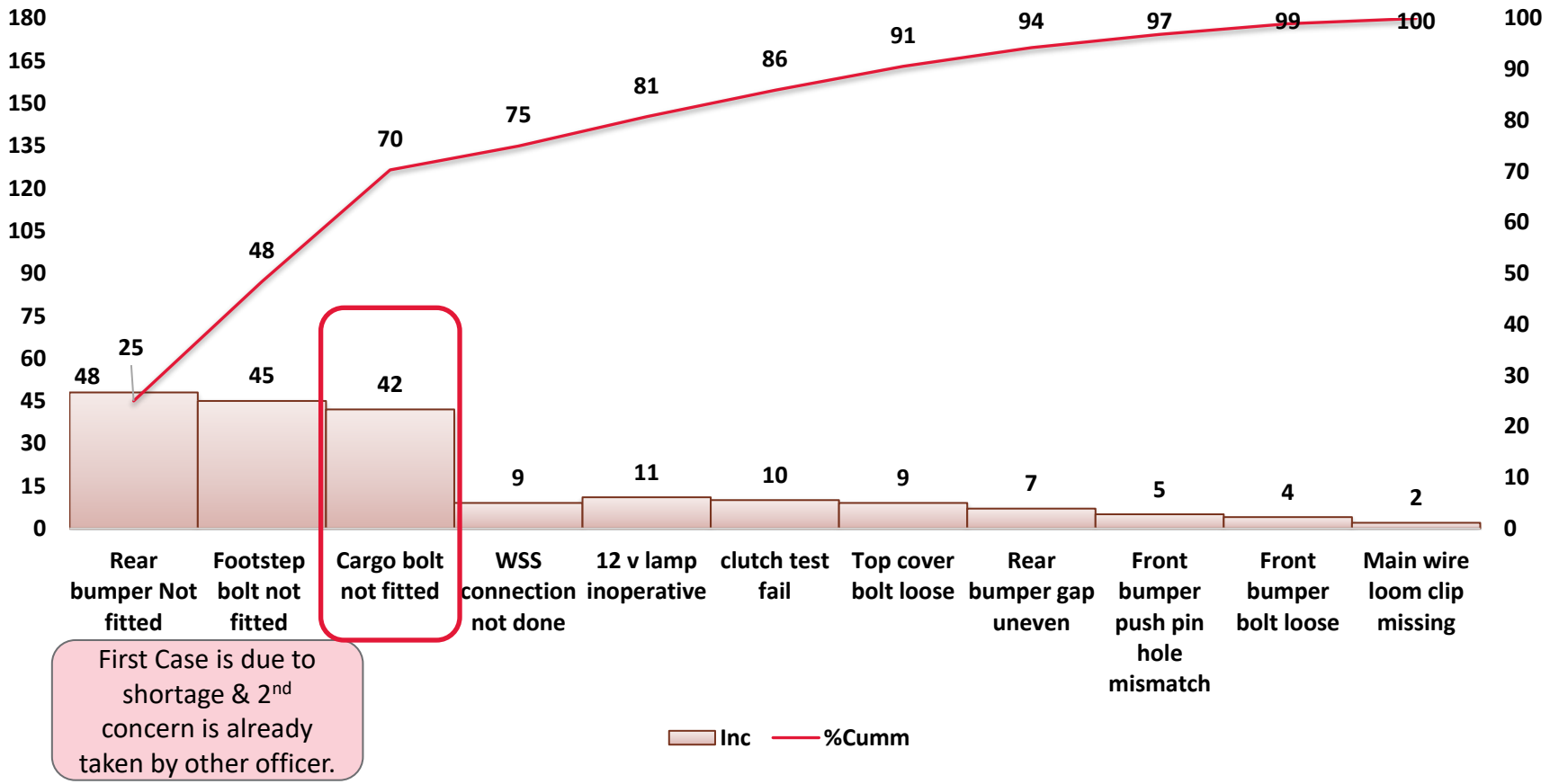
2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice

▪PARETO CHART ANALAYSIS FOR BWT 10 FINAL LINE– June 19 to Aug-19



First Case is due to shortage & 2nd concern is already taken by other officer.

Inference :- Cargo bolt not fitted is a chronic and time consuming defect at FAI .

2. Justify the choice (reasons)

Problem Criticality (Severity)

- **Importance & Severity of Concern :-**

:

- **Cargo bolt not fitted V1 concern.**
- **Major contributor in FRC on FAI**
- **Avg. 14 cases reported in month**
- **Skill operator required for rework.**
- **Rework activity is critical in offline activity**
- **Repair cost is very higher RS <u>24193</u> per year .**

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	25
Avg Concerns/Day	Nos	1.0
Total Required Rework time per day	Min	25
Effective working time	Min	465
Operator required /day	nos	0.05
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	80.65
Avg no. of working days	days	25
Operator cost /month	Rs	2016.1
Operator Cost/year	**Rs**	**24193.5**

2. Justify the choice (reasons)

Problem Criticality (Severity)

- Cargo bolt not fitted, if this defect occurs it is resulted into highly customer dissatisfaction and it may cause sever accident during driving . Severity of concern is 7 & 8

- Cargo bolt Cross fitted / loose/ not fitted contributes leads to major rework.

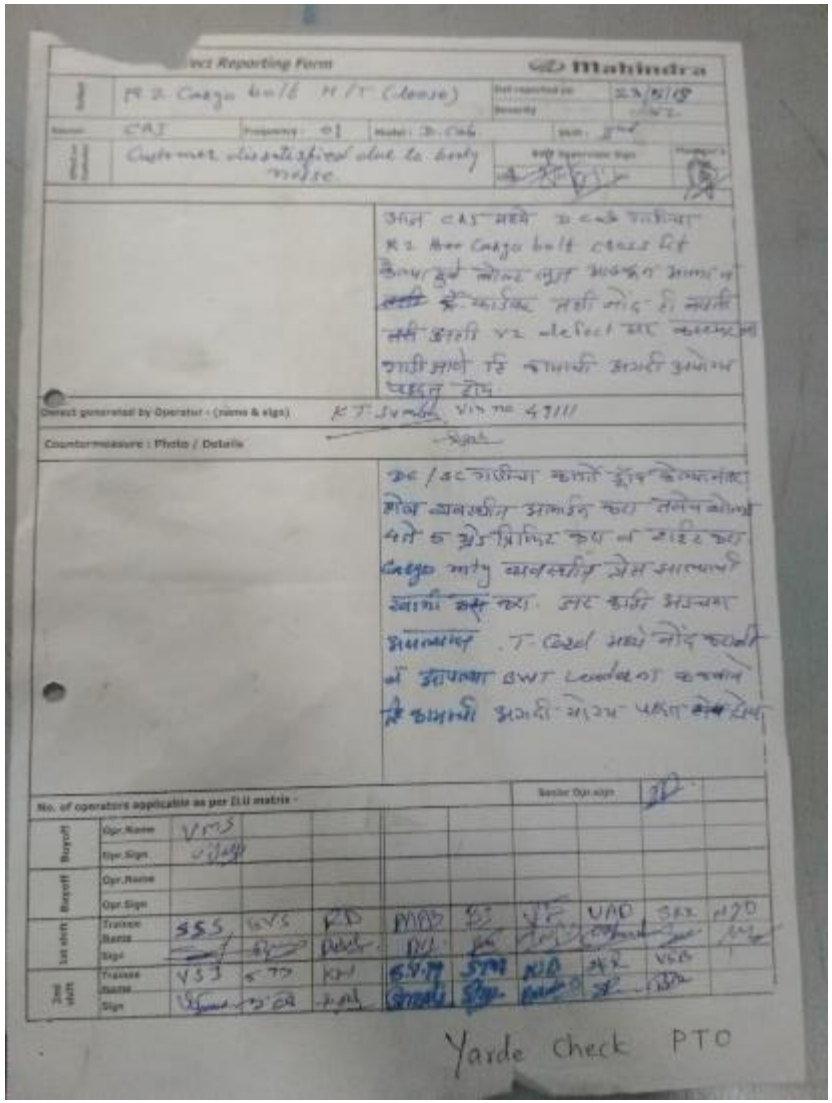
- It creates Unsafe condition to next stage operation.

- Require 1hrs to 2 hrs for rework the one vehicle.

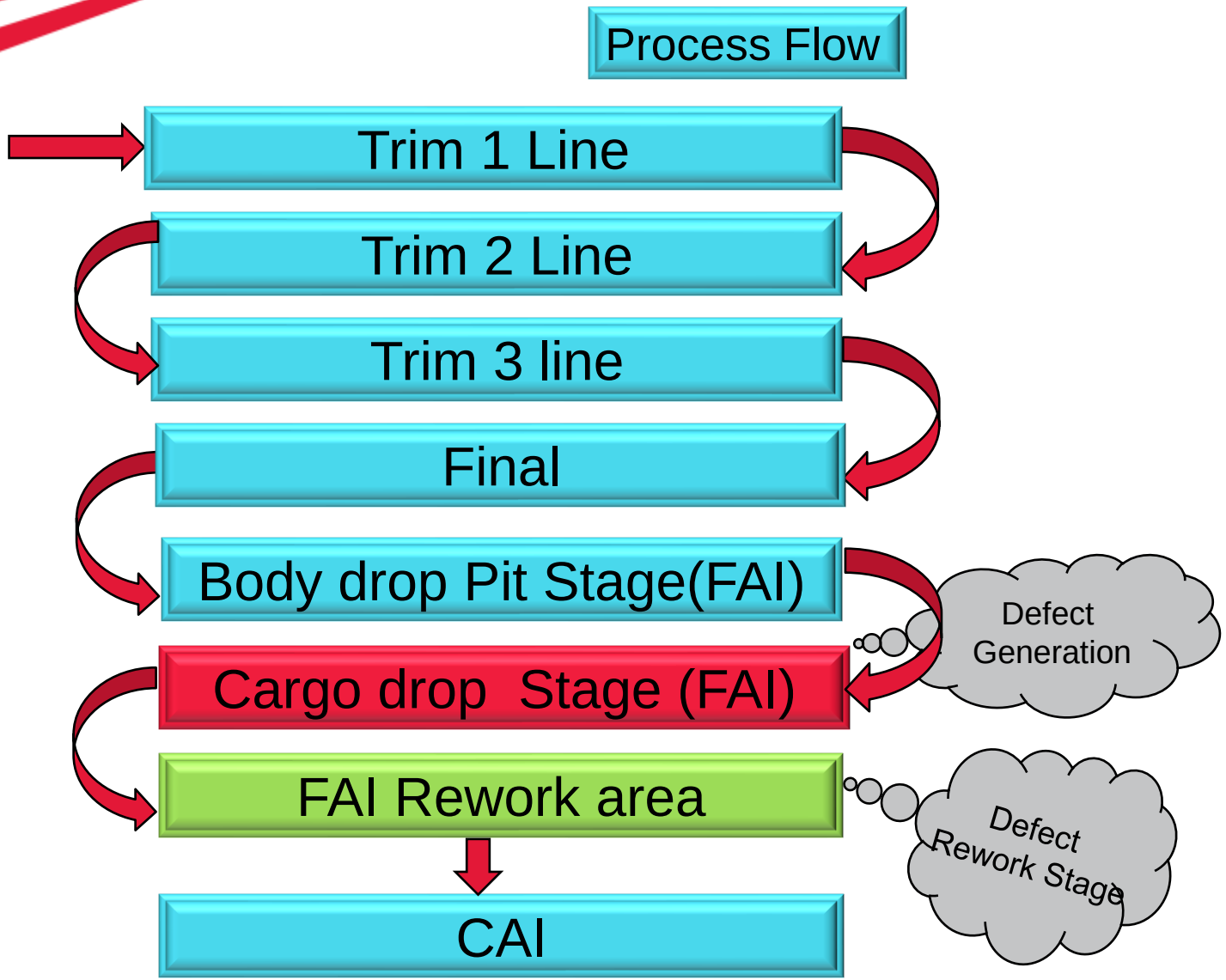
2.b Immediate Containment Action

- Vehicle reworked at FAI rework area for the defect Cargo bolt not fitted .

- Defect communicated to the stage operator through DRF.

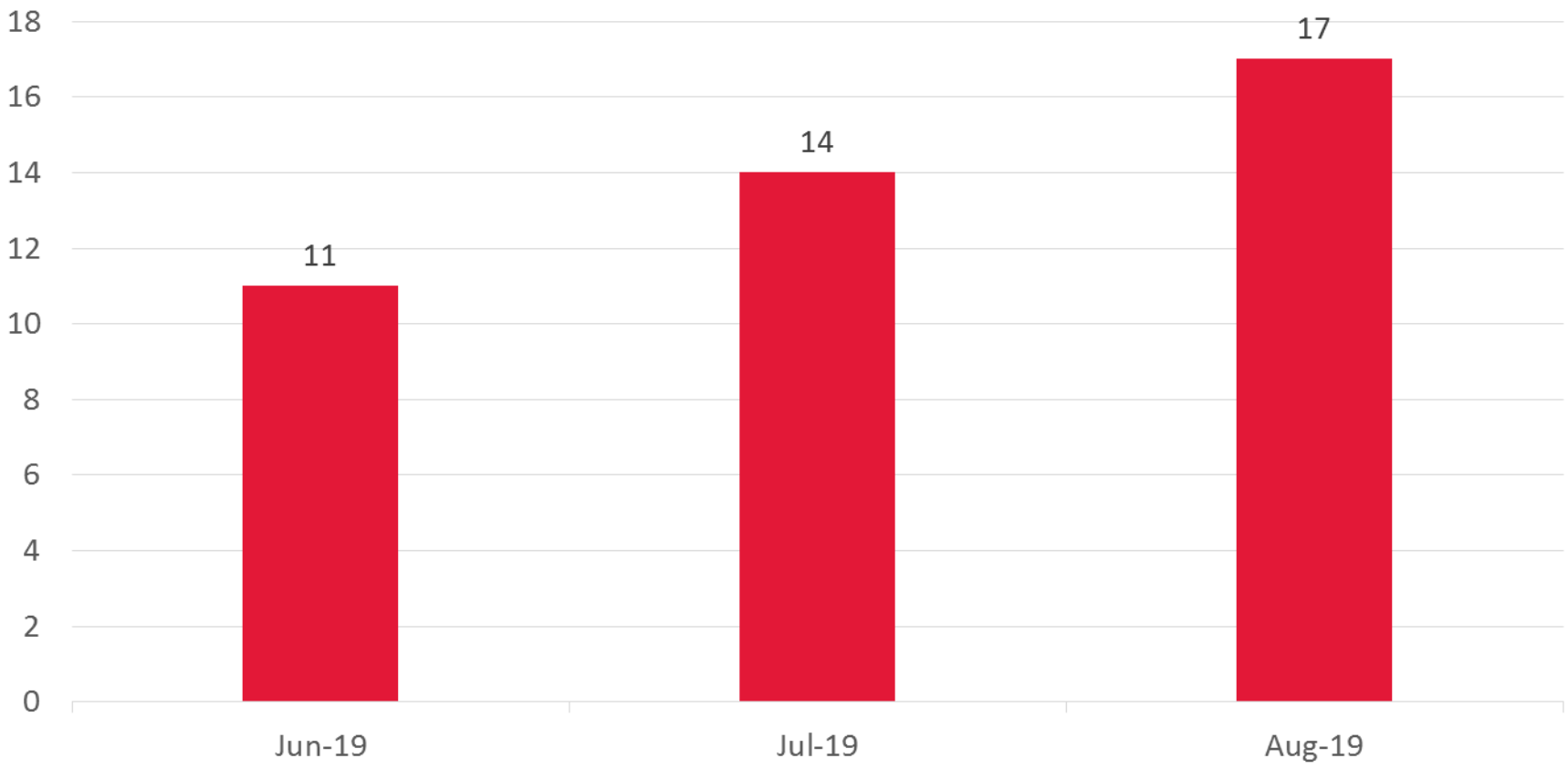


3. Understand the current situation



3. Understand the current situation

Month wise CARGO BOLT NOT FITTED concern Trend at final line for June19- to Aug-19



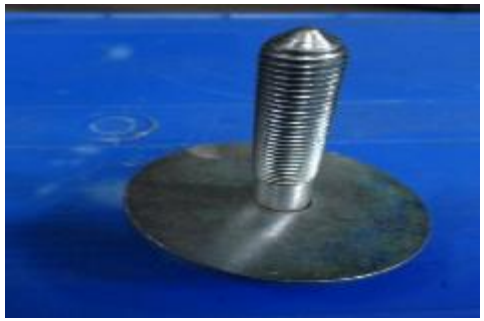
Inference – Month on month defect trend is increasing

3. Understand the current situation

Parts used For Cargo fitment Process.
(2 Types parts and 1 Types Hardware)

Cargo Mounting insulator .

Cargo mounting insulator Round.



Bolts .

Cargo

3. Understand the current situation

Operation.

Drop of Cargo body on chassis.

Pre fitment of Bolt on chassis cargo body area .

Fitment of bolt on chassis body.

3. Understand the current situation

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : Scorpio/Xylo TCF

Cell / BWT / Line: BWT 10

Requested By- MQA | Date -22.05.19 | Source -CAI | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement :CargoBolt not fitted

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- | Model : export S/C

Date of Incidence/ Assy : 14.5.19 | Shift / Time :2nd | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :

Vehicle reworked

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - | BWT . :

Does a control already exist ? ☐ YES ☐ NO

Stage / Post Analysis -

Operator Name : GT Bhadane,Vinod ingale , manoj bane

Man(Human Action)

Is operator trained ?	☑ YES ☐ NO
Operator level (LL.U)	L
OR Dexterity level(1…4)	3
Is operator under training ?	☐ YES ☑ NO
Is operator temporary for Stage ?	☐ YES ☑ NO
Bad rework ?	☐ YES ☑ NO

Method(process)

Main steps are respected ?	☑ YES ☐ NO
Is there a related key point in SOP / SOS?	☑ YES ☐ NO
Are the reasons for key point known ?	☑ YES ☐ NO
Are operating reqmnts respected ?	☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ?	☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ?- Amit Jadhav	☑ YES ☐ NO

Other

Is there a Control Plan ? (Record in Analysis table)	☑ YES ☐ NO
Is the packaging standard defined for the part ?	☑ YES ☐ NO
Is the packaging standard respected ?	☑ YES ☐ NO
Is there any alert mechanism for failure ?	☐ YES ☑ NO

Machine

Is the specified /Tool / Fixture facility used ?	☑ YES ☐ NO
Last calibration dates	
Facility /Tool / Fixture in good condition ?	☑ YES ☐ NO
Is it a substitute facility ?	☐ YES ☑ NO
Are machine parameters known ? (Record in Analysis table)	☑ YES ☐ NO
Parameters are respected ?	☑ YES ☐ NO

Product Failure

In only one shift ?	☑ YES ☐ NO
In only one line ?	☑ YES ☐ NO
The problem follows the vehicle ?	☐ YES ☑ NO
The problem follows the part ?	☑ YES ☐ NO
Is it a repeat problem ?	☑ YES ☐ NO
Is the suspected parameter measurable ? (Record in Analysis table)	☐ YES ☑ NO

Material/ Consumable

Is the specified product used ?	☑ YES ☐ NO
Did it evolve / Change ?	☐ YES ☑ NO
Is the storage SOP respected ?	☑ YES ☐ NO
Is the shelf life / expiry date respected ?	☑ YES ☐ NO
Is the part as per FIFO?	☑ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusionon Gemba (Attach Sketch/ Photograph if necessary) -
Vehicle reported for Cargo Bolt not fitted
History - This vehicle was rolled down on 14.05.19 2nd shift
Process flow - Cargo bolt is fitted at cargo drop stage .
Detection -1) At Final check point exist for cargo bolt fitment.
Traveler card - 1) No concern relateted to cargo bolt not fitted
Packaging - As per standared
Defect history - Cargo bolt cross fitted reported on line
Failed part :nil
During gemba analysis it is found that cargo hole get mismatch due to hole present on chassis bracket and in drawing given slot.Due to which hole get

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark

4M analysis Good and Not so good observations

MAN	Main step are respected by associate.
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Product failure	Problem follows part
Machine	Tools used are calibrated
Material	Handling and shelf life of part adhered

3. Understand the current situation

4M Analysis Observation.

- Concern Reported in Pit at wheel alignment stage.
- **All L3 Operators Working** on Stage.
- Concern Reported in SC/DC vehicle **all models .**
- Concern reported as Cargo bolt not fit /Cross fit .
- **Rework** is done at the rework area at pit area - **not possible in tact time on stage** .
- **Rework** Takes more than 1 **Hrs** time if it got slip .

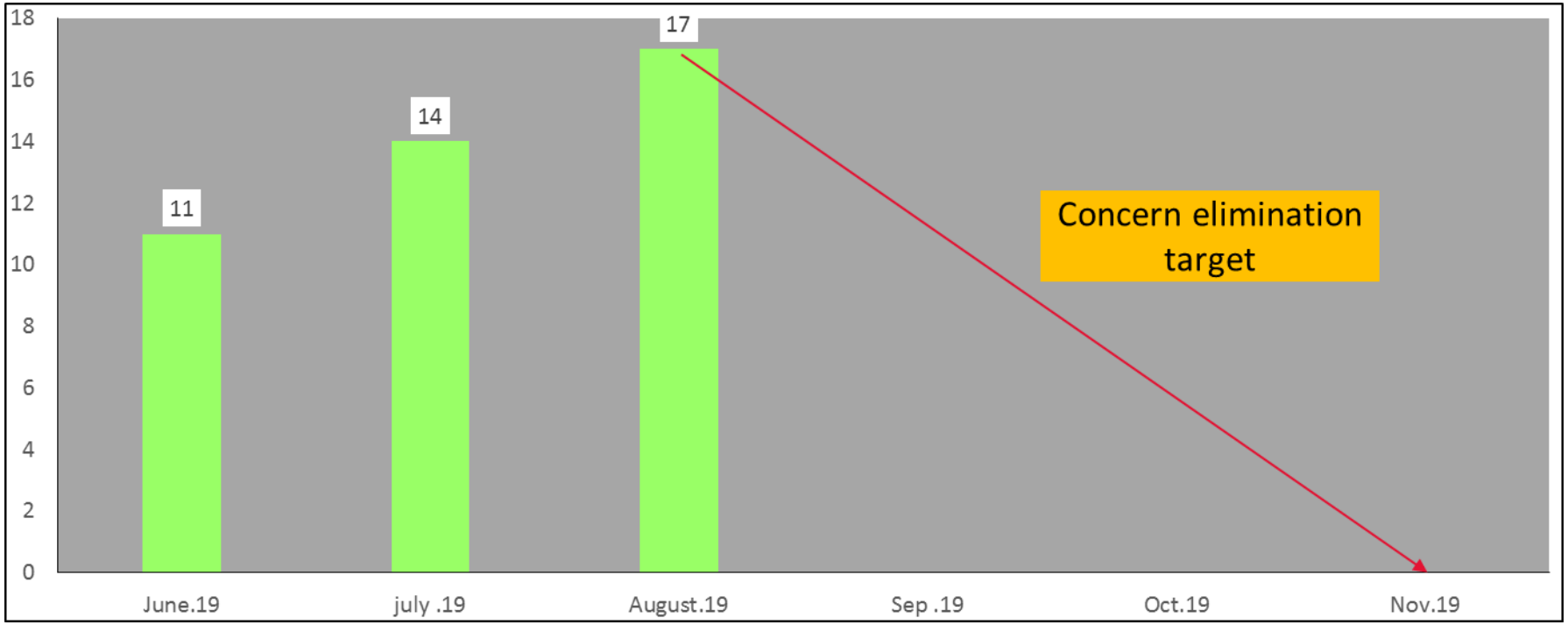
4. Select targets

Present Level:

Average 14 Defects/Month

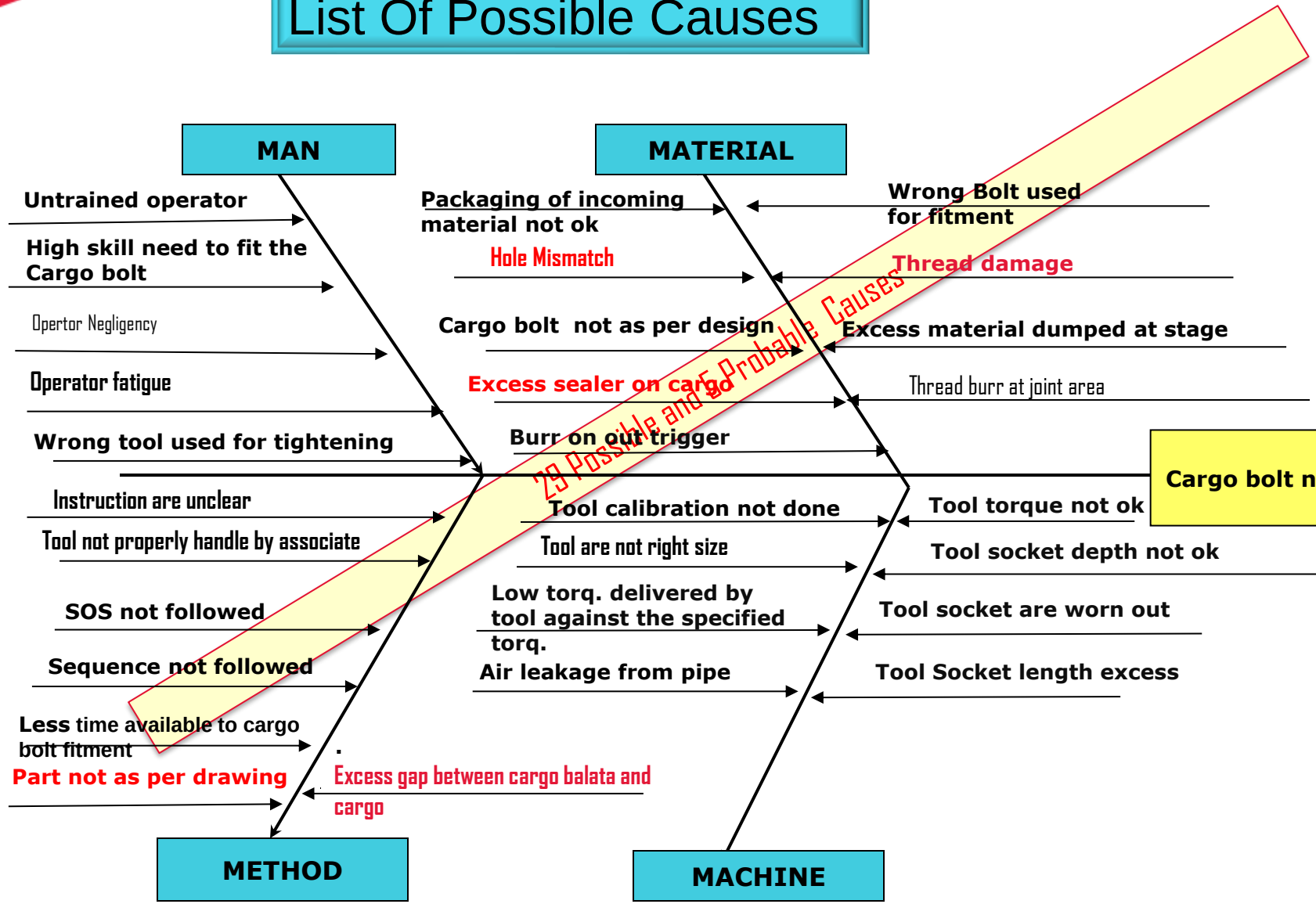
Target:

To Eliminate defect of Cargo bolt not fit by Nov 2019



Step 5. Analysis

List Of Possible Causes



5. Analysis

5.1 List of Possible causes: Brainstorming

Sr no	Possible cause	Nilesh	Sachin Bhambra	Santosh jadhav	Dilip junnare	Madhav hrhikar	Total
1	Untrained Operator	1	1	1	1	3	7
2	High skill need to fit cargo bolt	1	1	1	1	3	7
3	Operator fatigue	1	1	1	1	1	5
4	Operator Negligency	1	1	1	1	1	5
5	Wrong tool used for tightening	1	1	1	3	1	7
6	Packaging of incoming material not ok	1	1	1	1	1	5
7	Hole mismatch	9	9	9	3	9	39
8	Cargo bolt not as per design	3	3	3	3	3	15
9	Excess sealer on cargo	9	9	9	3	9	39
10	Thread damage	9	3	9	9	9	39
11	Burr on out trigger	3	3	3	3	3	15
12	Excess material dumped at stage	1	1	1	1	1	5
13	Thread burr at joint area	1	1	1	1	1	5
14	Wrong Bolt used for fitment	3	3	3	3	3	15
15	Instruction are unclear	1	1	1	1	1	5
16	Tool not properly handl by associate	1	1	3	3	1	9
17	SOS not followed	3	3	3	5	3	17
18	sequence not followed	3	3	3	5	3	17
19	Less time available to cargo bolt fitment	3	3	3	3	3	15
20	Part not as per drawing	9	9	9	9	9	45
21	Excess gap between cargo balata and cargo	9	9	9	9	9	45
22	Tool calibration not done	3	3	3	3	1	13
23	Tool are not right size	3	3	3	3	3	15
24	Low torq. delivered by tool against the specified torq.	3	3	3	3	3	15
25	Air leakage from pipe	3	3	3	3	3	15
26	Tool torque not ok	3	3	3	3	3	15
28	Tool Socket depth not ok	3	3	3	3	3	15
29	Tool socket wear out	3	3	3	3	3	15

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 5- Significant impact.
Threshold Value for selecting probable cause is 39.

Step 5. Analysis

5.1 List of Probable causes: Brainstorming

List of causes not considered as probable causes and reasons for not considering

Sr nos	Possible cause	Reasons
1	Untrained Operator	Trained operator deploy on stage
2	High skill need to fit the cargo bolt	L3 Dexterity level deploy on stage
3	Operator fatigue	No Operator fatigue observe for fitment
4	Operator Negligency	No operator negligency observe
5	Wrong tool used for tightening	Tool used as per specification
6	Packaging of incoming material not ok	Packaging as per packaging standared
8	Cargo bolt not as per design	cargo bolt found ok as per design
11	Burr on out trigger	Part found as per specification
12	Excess material dumped at stage	Two bin system available at stage
13	Thread burr at joint area	Joint found ok
14	Wrong Bolt used for fitment	Bolt found as per specification and drawing
15	Instruction are unclear	Instruction clear mentioned in SOS
16	Tool not properly handl by associate	Tool handling found ok
17	SOS not followed	SOS followed by associate
18	sequence not followed	Sequence followed by associate
19	Less time available to cargo bolt fitment	Sufficient time allocate to complete work
22	Tool calibration not done	Tool calibration done as per scheduled
23	Tool are not right size	Tool as per sepecification
24	Low torq. delivered by tool against the specified torq.	Torque found as per given specification
25	Air leakage from pipe	No air leakage observe from pipe
26	Tool torque not ok	Tool torque found ok
28	Tool Socket depth not ok	Tool socket depth found ok
29	Tool socket wear out	No socket wear out observe

5. Analysis

5.1 List of Probable causes: Brainstorming

Sr No	Possible cause
7	Hole mismatch
9	Excess sealer on cargo
10	Thread damage
20	Part not as per drawing
21	Excess gap between cargo balata and cargo

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Hole mismatch	1) Visual 2) During bolt pre fitment	CFT member observe the stage for 2 hrs. Observations are as below... 1. Cargo and Chassis hole not concentric with each other . .	Hypothesis valid

20 nos of part check for Hole mismatch out of which 4 found not ok

Sr no	Model	Accessibility
1	SC	ok
2	DC	ok
3	SC	ok
4	SC	Not ok
5	DC	ok
6	SC	ok
7	SC	ok
8	SC	Not ok
9	DC	ok
10	DC	ok
11	DC	ok
12	DC	ok
13	DC	ok
14	SC	ok
15	SC	ok
16	DC	ok
17	SC	Not ok
18	DC	ok
19	SC	Not ok
20	SC	ok

Concern	Cargo hole position ok	Cargo hole position not ok
Vehicle Checked		

At body drop under pit it is observe that cargo hole get mismatch at time cargo bolt fitment .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	Excess sealer on cargo	Visual inspection	20 Nos of vehicle check for Sealer and it is found that 3 nos found Sealer at cargo bolt tightening area .	Hypothesis valid

20 nos of cargo check for excess sealer out of which 3 nos found excess sealer

Concern	No sealer on cargo	Excess sealer on cargo
Vehicle Checked	|||||	|||

Sr No	Model	Sealer applicabilit	Remark
1	SC	ok	
2	DC	ok	
3	SC	not ok	Sealer found at cargo cross mbr at bolt tightning area
4	SC	ok	
5	SC	ok	
6	DC	ok	
7	DC	ok	
8	DC	ok	
9	DC	ok	
10	DC	ok	
11	SC	not ok	Sealer found at cargo cross mbr at bolt tightning area
12	SC	ok	
13	SC	ok	
14	SC	ok	
15	SC	ok	
16	DC	ok	
17	SC	not ok	Sealer found at cargo cross mbr at bolt tightning area
18	DC	ok	
19	SC	ok	
20	DC	ok	

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
10	Thread damage	Ring gauge	50 Nos. Bolt check and found thread ok	Hypothesis Invalid

50 nos of bolt check for thread damage Thread found ok

Sr no	Thread check	Remark
1	found ok	
2	found ok	
3	found ok	
4	found ok	
5	found ok	
6	found ok	
7	found ok	
8	found ok	
9	found ok	
10	found ok	
11	found ok	
12	found ok	
13	found ok	
14	found ok	
15	found ok	
16	found ok	
17	found ok	
18	found ok	
19	found ok	
20	found ok	
21	found ok	
22	found ok	
23	found ok	
24	found ok	
25	found ok	

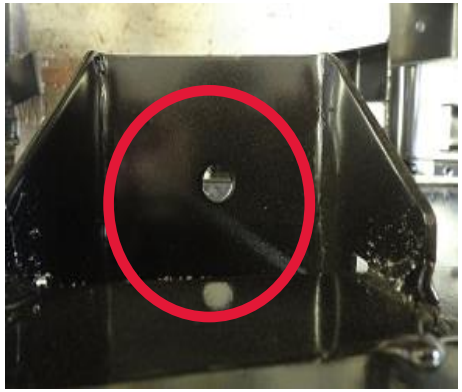
Sr no	Thread check	Remark
26	found ok	
27	found ok	
28	found ok	
29	found ok	
30	found ok	
31	found ok	
32	found ok	
33	found ok	
34	found ok	
35	found ok	
36	found ok	
37	found ok	
38	found ok	
39	found ok	
40	found ok	
41	found ok	
42	found ok	
43	found ok	
44	found ok	
45	found ok	
46	found ok	
47	found ok	
48	found ok	
49	found ok	
50	found ok	

TEST OF HYPOTHESIS

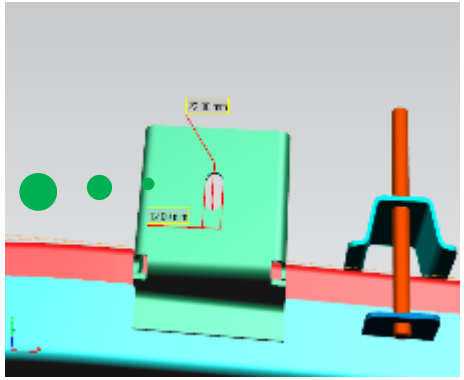
5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
20	Part not as per drawing	Drawing	During observation it is observe that In drawing chassis outrigger found as slot but in actual part found with circular hole	Hypothesis valid

10 nos of part check as per drawing during which it is observe that part not as per drawing



Slot available on chassis outrigger as per drawing



Sr no	Model	part as per drawing	Remark
1	SC	Not ok	12 mm hole found on chassis outrigger
2	DC	Not ok	
3	SC	Not ok	
4	SC	Not ok	
5	DC	Not ok	
6	SC	Not ok	
7	SC	Not ok	
8	SC	Not ok	
9	DC	Not ok	
10	DC	Not ok	

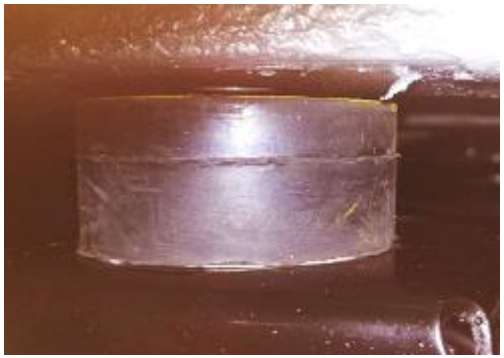
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
21	Excess gap between cargo balata and cargo .	1) Tapper scale . 2) Visual	During observation it is found that Outrigger position found not ok in Z direction	Hypothesis valid

20 nos of vehicle checked for gap ,it is observe that 4 nos vehicle found excess gap

Concern	No gap	Excess gap
Vehicle Checked		



Sr no	Model	Gap between balata and cargo	remark
1	SC	ok	
2	DC	ok	
3	SC	ok	
4	SC	Not ok	8mm gap observe
5	DC	ok	
6	SC	ok	
7	SC	ok	
8	SC	Not ok	5 mm gap observe
9	DC	ok	
10	DC	Not ok	10 mm gap observe
11	DC	ok	
12	DC	Not ok	12 mm gap observe
13	DC	ok	
14	SC	ok	
15	SC	ok	
16	DC	ok	
17	SC	Not ok	7 mm gap observe
18	DC	ok	
19	SC	Not ok	10 mm gap observe
20	SC	Not ok	7 mm gap observe

Observation :- Due to excess gap between mounting and cargo cross mbr cargo bolt is not prefitted .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Hole mismatch	Visual observations	CFT member observe the stage for 2 hrs. Observations are as below… 1.Cargo and Chassis hole not concentric with each other (Tolerance stack up not allowing to Cargo nut position to match with Chassis hole position)	**Hypothesis valid**
2	Excess sealer on cargo	Visual	20 Nos of vehicle check for Sealer and it is found that sealer observe for 3 cargo at cargo bolt tightening area .	**Hypothesis valid**
3	Part not as per drawing	Drawing Checking	In drawing chassis outrigger found as slot but in actual part found with circular hole .	**Hypothesis valid**
4	Excess gap between cargo balata and cargo .	1) Tapper scale . 2) Visual	During observation it is found that Outrigger position found not ok in Z direction	**Hypothesis valid**

4 Valid Causes

5.3 Identify the root causes:

WHY - WHY Analysis

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Hole mismatch	Cargo and Chassis hole not concentric with each other	Both parts as with in specifications till issue is there	Tolerance stack up not allowing to Cargo nut position to match with Chassis hole position.	Change the design
2	Excess sealer on cargo	Sealer application observe on mounting area from paint shop	Variation in sealer application by associate	No awareness at paint shop to avoid sealer application in cargo bolt mounting area	Provide training / Communication
3	Part not as per drawing	In drawing chassis outrigger found as slot but in actual part found with circular hole	Part supply found as per old drg	--	Part to be corrected
4	Excess gap between cargo balata and cargo .	Cargo surface not matching with chassis bkt	Chassis Outrigger position found shifted in Z direction	Variation in supplier welding fixture	Correct the fixture.

6. Implement the corrective measures

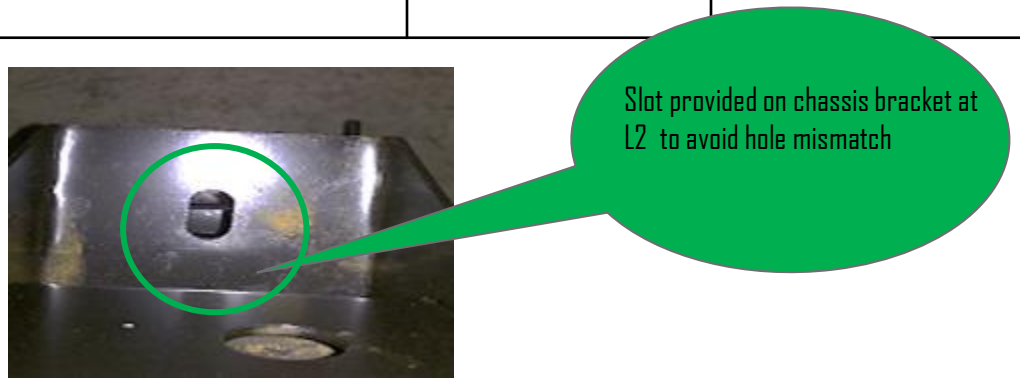
6.1 Actions Decided

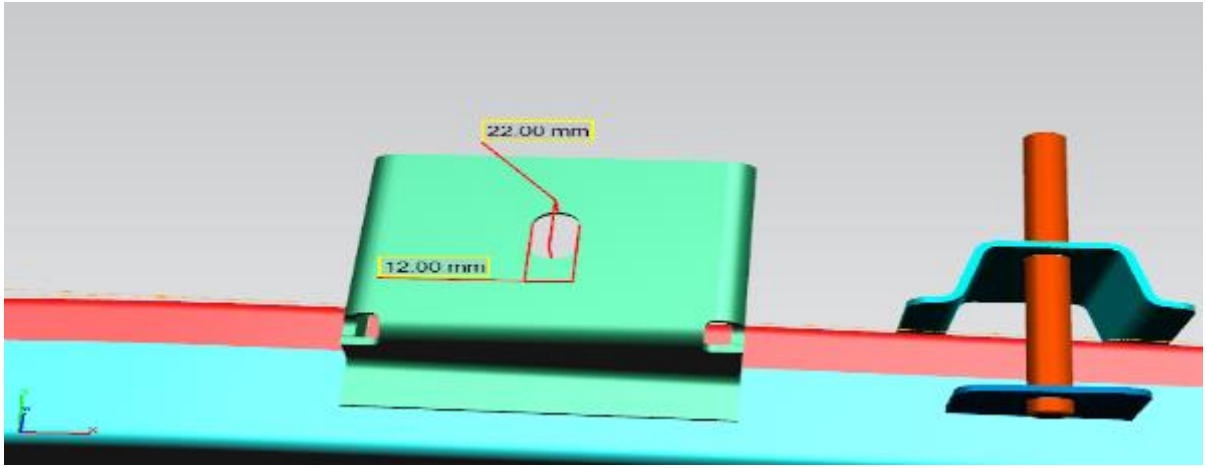
No	Root Cause	Action plan	Target date	Responsibility	Status
1	Tolerance stack up not allowing to Cargo nut position to match with Chassis hole position.	Change the design	12.7.19	Santosh jadhav	Closed
2	No awareness at paint shop to avoid sealer application in cargo bolt mounting area	Provide training / Communication	18.10.19	Anant Kulkarni	Closed
3	Part supply found as per old drg	Part to be corrected	12.7.19	Madhav Hardikar	Closed
4	Variation in supplier welding fixture	Correct the fixture.	10.8.19	Satish Pagar	Closed

6. Implement the corrective measures

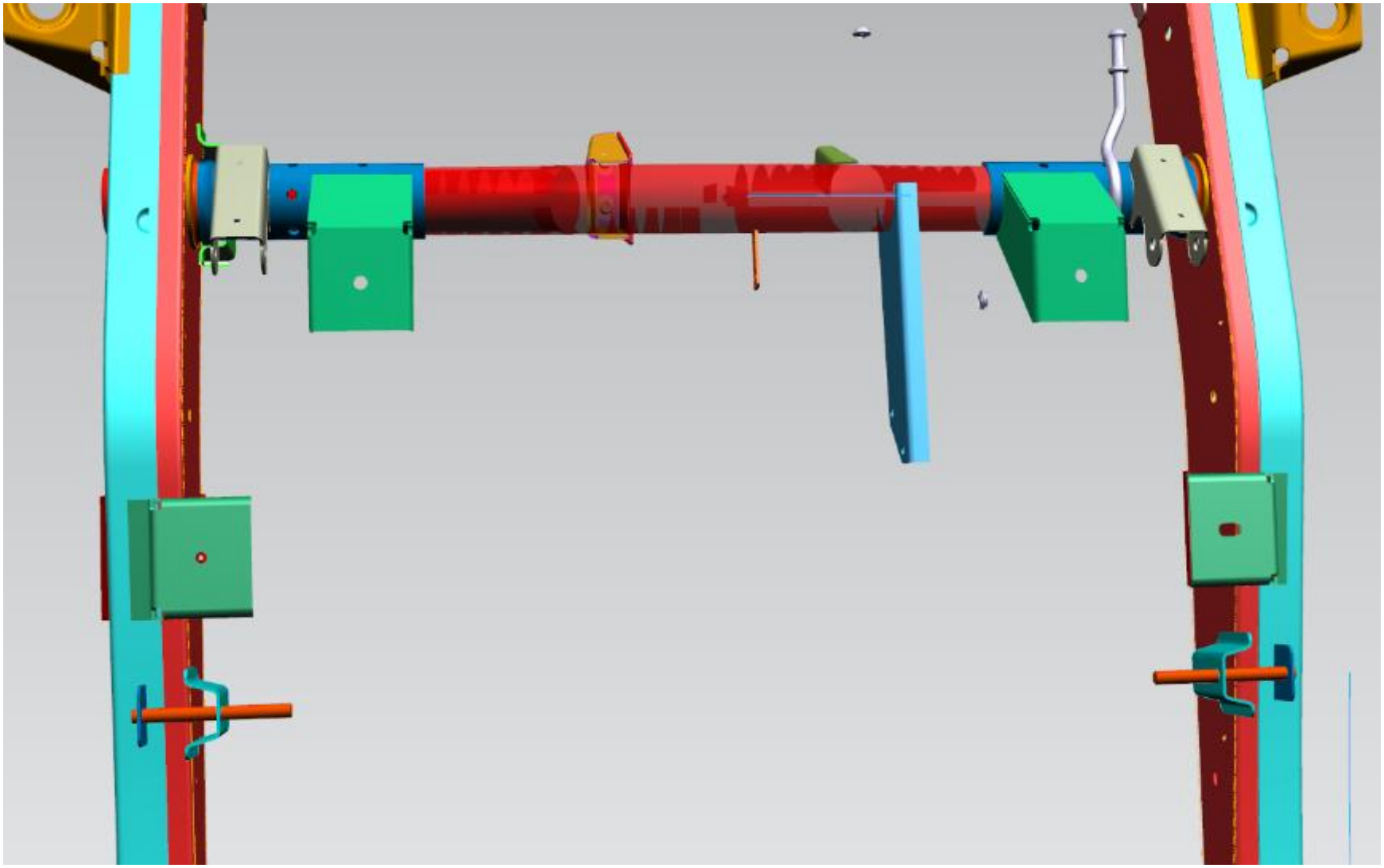
No	Root Cause	Action plan	Target date	Responsibility	Status
1	Hole Mismatch	As per stack up Slot Provided at Chassis bracket L2 outrigger	12.7.19	Santosh Jadhav	Closed

Sr no	Trial taken	Remark
1	found ok	
2	found ok	
3	found ok	
4	found ok	
5	found ok	
6	found ok	
7	found ok	
8	found ok	
9	found ok	
10	found ok	



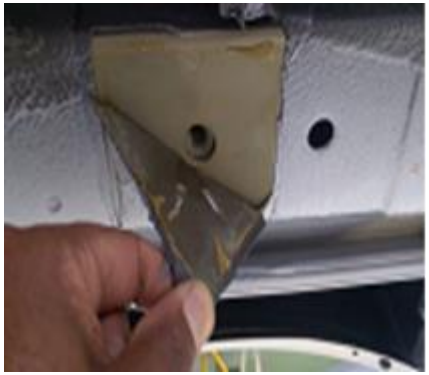


Drawing



6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
2	Excess sealer on cargo	Excess Sealer application avoided at mounting area by providing patches at cargo mounting area	18.10.19	Anant Kulkarni	Closed



Excess Sealer application at mounting area

Before

Excess Sealer application avoided at mounting area by providing patches at cargo mounting area

After

6. Implement the corrective measures

Gemba Kaizen Sheet

Plant :Nashik / Plant-I	PU / Shop / Dept. :Paint/MQS/sealer Line	Cell / Circle : Sealer
Machine / Equipment / Workstation No. :	Machine / Equipment / Workstation Name :	18.10.19

Pillar Applicability (pl. tick) ↓

KK / TM	JH	PM	QM	DM	E&T	OTPM	SHE
☐	■	☐	■	☐	☐	☐	☐

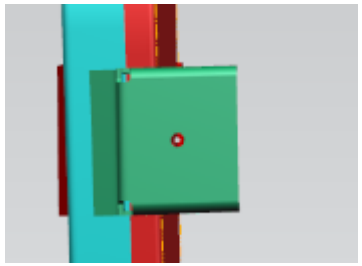
Theme :- No hole block by UBS spray on carogo

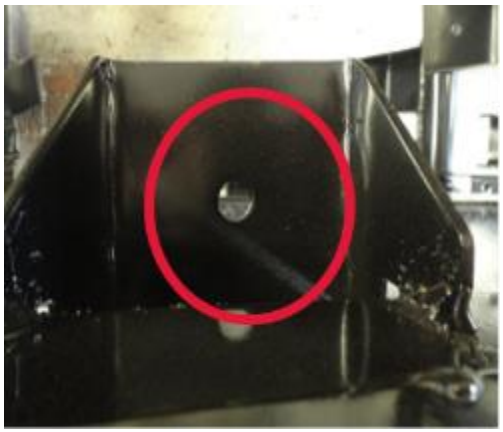
BEFORE	AFTER
Description in details :-	Description in details :-
UBS on Chassis mounting hole due to self falling masking not stick proper in place	Before masking, mark by fixture & stick proper self falling in place

Sr.no.	Description	Benefits
1	Before masking, mark by fixture & stick proper self falling in place	TCF Concern eliminate
2		Quality improved

6. Implement the corrective measures

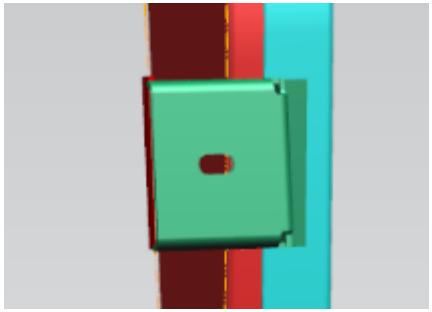
No	Root Cause	Action plan	Target date	Responsibility	Status
3	Part not as per drawing	Part corrected as per drawing – Slot provided at L2 and R2 cargo bolt fitment area.	12.7.19	Madhav hardikar	Closed





1) Hole on chassis bracket
2) As per drawing slot present on chassis member L2 and R2.

Before



1) Slot provided on chassis bracket
2) Drawing modification done for R2 and L2 bracket on chassis member

After

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
4	Excess gap between cargo balata and cargo.	Outrigger position corrected in Z direction at chassis .	10.8.19	Satish Pagar	Closed

Outrigger position corrected in Z direction at chassis

No gap between cargo balata and cargo .

6. Implement the corrective measures

Chassis Supplier SCAP Report

TO BE FILLED BY M&M

M & M Ltd. Automotive Sector	SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP)	PU Name : Sr.No: Date:19/07/2019

Supplier:- KLT AUTOMOBILE & TUBULAR PRODUCTS LTD — Supplier Code: DK055
Part Description: FRAME ASSY 4WD DC — Part No.: 0201AAK10520N
When Observed (Date / Period) : 19/07/2019 — GR/Invoice No.:

A PROBLEM STATEMENT Should Answer — ISSUE REPORTED : REPEAT ☐ / 1'ST TIME ☑

What: Excess gap between cargo and chassis mtg
Who: M&M Nashik
Where:
When: Date: 19/07/2019
Why:
How much Qty: Qty-01 Nos

(Problem statement should answer 5W 1H is What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)

5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
It was not detected at Final Inspection stage	No Inspection point in check list	No checking method available		

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:

1) All stock of chassis checked at M&M end. 7 chassis found defective which was hold at end will send back

2) All stock checked at KLT end and defective parts reworked

L ROOT CAUSE ANALYSIS OF PROBLEM: ☑ YES ☐ NO

Machine
Is the specified M/C used ? ☑ YES ☐ NO
Is M/C in good condition ? ☑ YES ☐ NO
Is it a substitute M/C ? ☐ YES ☑ NO
Are M/C parameters known ? ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

Man
Is operator trained ? ☐ YES ☑ NO
Is operator is under training ? ☑ YES ☐ NO
Is operator temporary ? ☐ YES ☑ NO
Is it due to Bad Rework ? ☐ YES ☑ NO

Measurement
Are instruments / gauges calibrated ? ☑ YES ☐ NO
Are Control Methods Capable to detect Non conformity? ☑ YES ☐ NO
Are specified Control Methods used ☑ YES ☐ NO

Excess gap between cargo and chassis mtg

Method
Are Control Plan/SOP available near M/C? ☑ YES ☐ NO
Are Control Plan respected ? ☑ YES ☐ NO
Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO

Material / Tooling
Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO
Is the Raw Matl /Tooling Supplier change ? ☐ YES ☑ NO
Is the heat treatment method change ? ☐ YES ☑ NO

Environment
Is process needs environmental control? ☐ YES ☑ NO
Is it implemented ? ☐ YES ☑ NO
Does it conform ? ☐ YES ☑ NO

Page 1

Analysis conclusion:-
After analysis it observed that Spare wheel cross member bow observed upto 10.0 mm due to spare wheel child part bow and welding distortion after welding spare wheel Winch bracket spare wheel support.

F CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	After analysis it observed that Spare wheel cross member bow observed upto 10.0 mm due to spare wheel child part bow and welding distortion after welding spare wheel Winch bracket spare wheel support.	1) Increase pressure in bending operation for spare wheel cross member flatness. 2) Spare wheel top clamp pressure pad changes and welding sequences changes. 3) Flatness checking point added in final inspection check list	Amit T	22/07/2019	22/07/2019

G ACTION TAKEN TO PREVENT RECURRENCE

Flat Square tool for checking spare wheel flatness

Station no. 8

No Gap found

Page 2

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT (HORIZONTAL DEPLOYMENT)

I Following quality documents changed due to implemented action plan :
(mention doc. no.)

Process Flow Chart	☐	Gauge Drawing	☐
PFMEA	☐	Fixture Drawing	☐
Control Plan	☐		
Work instruction / SOP	☐		

L EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem	☐ YES	☐ NO
Are actions planned completed / implemented	☐ YES	☐ NO
Is problem repeated in lots received after SCAP issue Date	☐ YES	☐ NO

FOR MAHINDRA & MAHINDRA (A.S.) — Signature / Date

SUPPLIER'S REPRESENTATIVE
Yogeshwar Vengurlekar
Signature / Date

MQS/SCAP/ F/01 — Rev.02

7. Check & confirm the results & effects

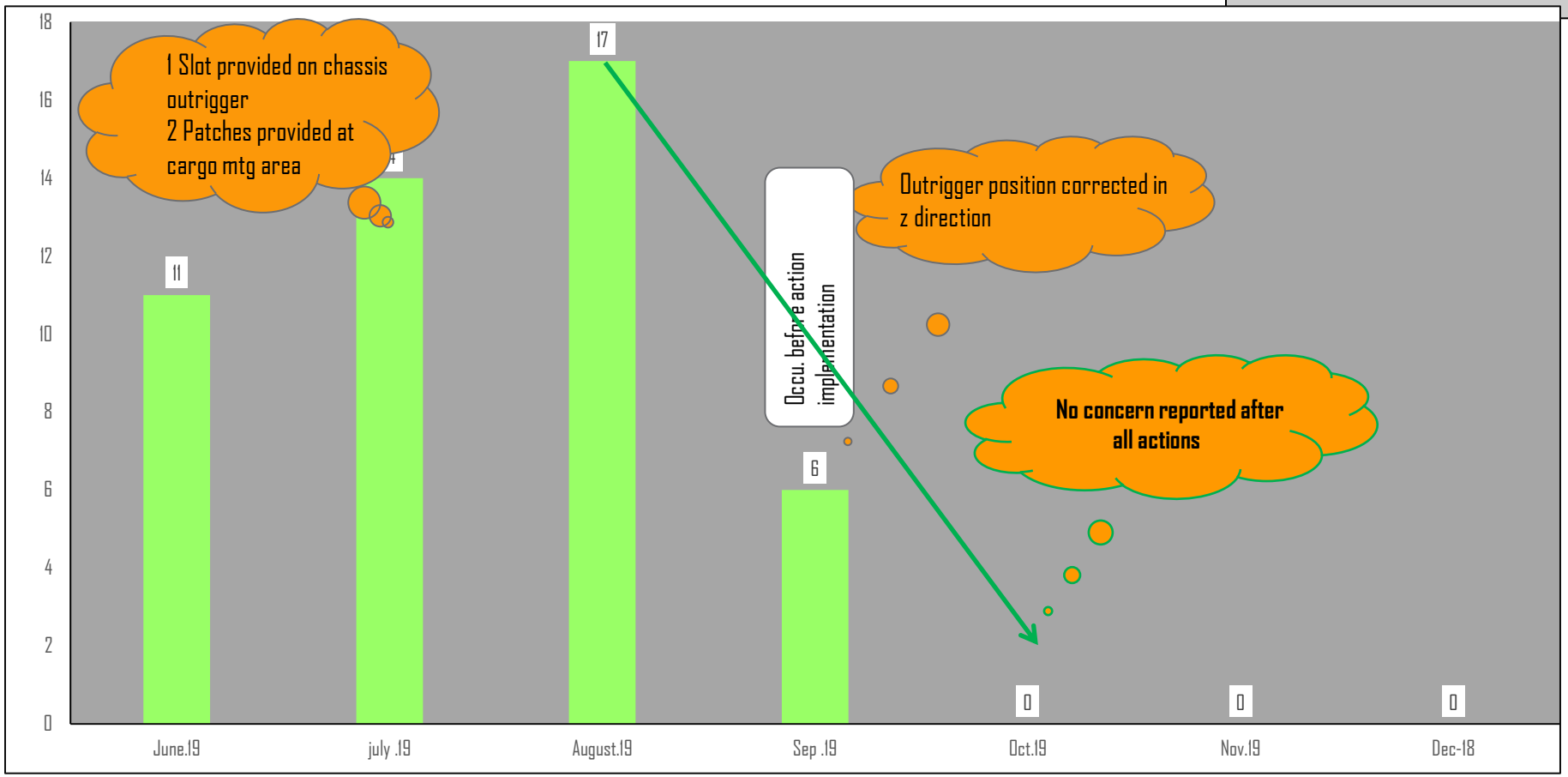
7.1 Verify Results:

S N	Action Taken	Testing Method	Results Before Action	Results After Action
1	As per stack up Slot Provided at Chassis bracket L2 outrigger	Verifying of part with drawing	Hole mismatch due to which cargo bolt not fitted .	Hole get align and due to which bolt get easily prefited and tightning.
2	Excess Sealer application avoided at mounting area by providing patches at cargo mounting area.	Visual	Due to excess sealer Cargo bolt get not fitted.	No issue observe for cargo bolt pre fitment and tightening .
3	Part corrected as per drawing – Slot provided at L2 and R2 cargo bolt fitment area.	Drawing verification .	Cargo bolt cross fitted and not fitted issue observe .	Cargo bolt get easily fitted due to
4	Outrigger position corrected in Z direction at chassis	Taper scale Visual checking	Excess gap observe between outrigger and balata .Pre fitment of cargo bolt not possible	Gap eliminated and cargo bolt get easily prefitted.

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend

X Axis- Month
Y Axis- cargo bolt not fitted .



Inference - Concern is eliminated after all 3 actions implemented.

1. Hole mismatch avoided.
2.Slot provided on chassis bracket.
3 Drawing changed

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	As per stack up Slot Provided at Chassis bracket L2 outrigger	12.7.19	Santosh Jadhav	Closed
2	Excess Sealer application avoided at mounting area by providing patches at cargo mounting area.	18.10.19	Anant Kulkarni	Closed
3	Part corrected as per drawing – Slot provided at L2 and R2 cargo bolt fitment area. Outrigger position corrected in Z direction at chassis	12.7.19	Madhav Hardikar.	Closed
4		10.8.19	Satish Pagar	Closed

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	Part Drawing	12.7.19	Santosh Jadhav	Closed
2	SOS FMEA (paint shop)	18.10.19	Eknath Kulkarni	Closed
3	Chassis Drawing modification.	12.7.19	Madhav Hardikar	Closed
4	Scap report.	10.8.19	Satish Pagar	Closed

Check & confirm the results & effects (Cont..)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) FRC and FBO Improved by 0.08 %
2) COPQ (Material cost+ Labour cost+ Rework cost) =(Apprx. Rs.24183 Per annum)is Saved

In tangible Benefits

1) Knowledge of Worker is Improved.
2) Improved Customer Satisfaction.

FILE MESSAGE

Ignore Junk Delete Reply Reply All Forward Meeting More Move to: ? Team Email Reply & Delete

Delete Respond

Wed 15-01-2020 09:03

DHANOKAR RAJESH

Re:MYB cost aproval

To DEORE NILESH; DHANOKAR RAJESH

okay approved

Sent from my Redmi Note 4X
On DEORE NILESH <DEORE.NILESH@mahindra.com>, 15-Jan-2020 8:22 AM wrote:
Dear sir,
Please check cost estimation for rework cargo bolt .
Required for MYB please check and confirm

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	25
Avg Concerns/Day	Nos	1.0
Total Required Rework time per day	Min	25
Effective working time	Min	465
Operator required /day	nos	0.05
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	80.65
Avg no. of working days	days	25
Operator cost /month	Rs	2016.1
Operator Cost/year	Rs	24193.5

Regards,
Nilesh

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Target date	Responsibility	Status
1	Drawing updation	12.7.19	Santosh Jadhav	Closed
2	QA matrix updation	18.10.19	Nilesh deore	Closed

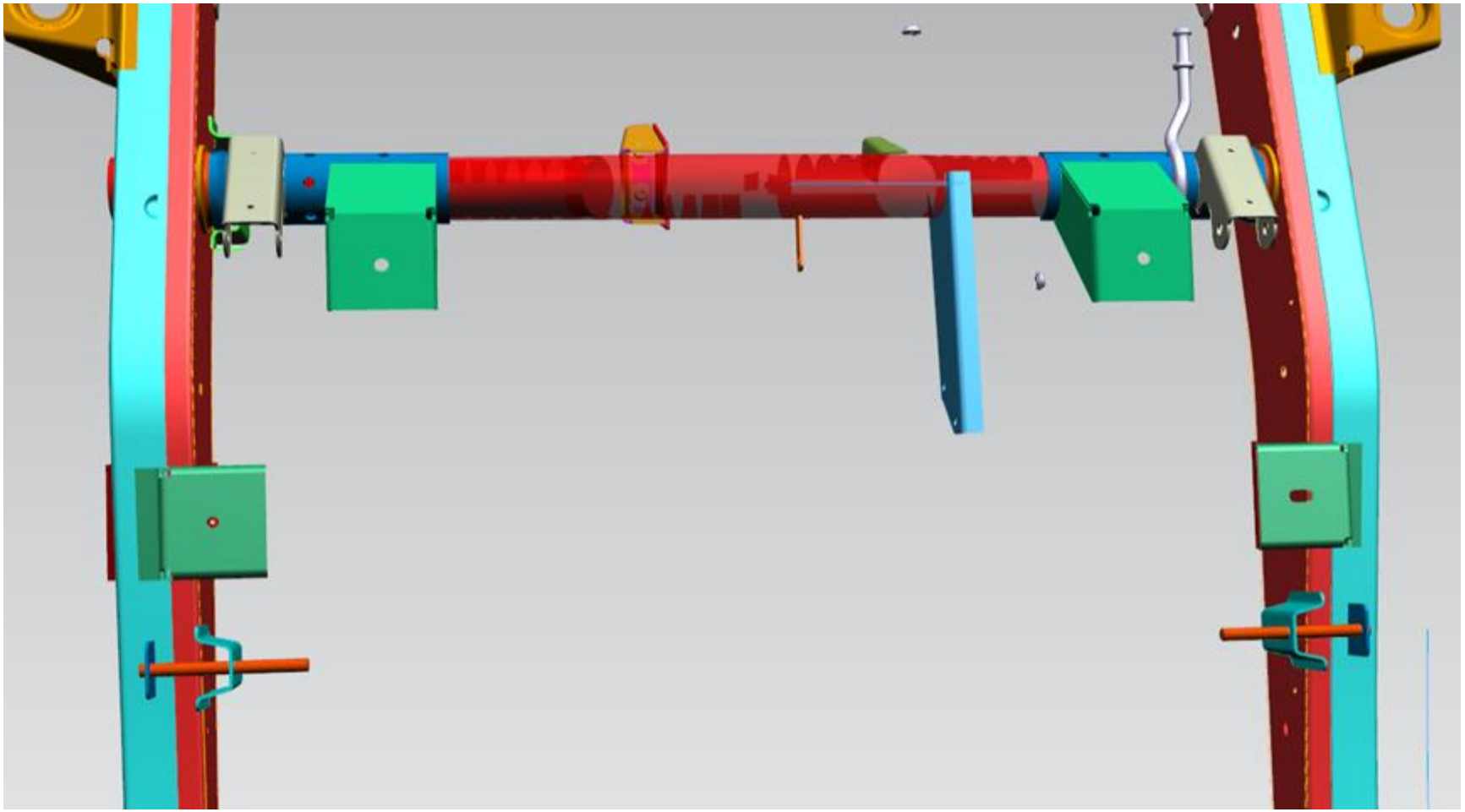
8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	On job training given to Cargo drop group and Roll down under pit Associate as per new Drawing updation.	25.5.19	Nilesh Deore	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

DRAWING



8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

QA Matrix

QA MATRIX

PU NAME	SCORPIO	INITIALIZATION DATE	06.04.2016	Global Quality Guarantee level.	
DEPT NAME	SXTCF	SIGNATURES			
SECTION NAME	FAI	Dept Head	Rahul Narkar	Section	80
		Module Manager	Taj Shaikh	Plant	99.6
		Section Manager	1 2 3 4	Revision date	18.10.19

Workstation No.	Process description (POS)	Date	Failure modes	Severity	Prev. occurence	Potential decfect	Inspection process: Section: BODY DROP	cargo	CONSOLE	Production activity	FILLING	Nitrogen testing	AC Filling	FLASHING	SEAT FIT	Plenum	UNDERPIT	BUYOFF	ROLL DOWN	ALLIGNMENT	Outside rework	Outside section: TRIM	CHASSIS	ROLLER	TESTER	SHOWER	WRAP	S & R	CAI	ECOS	Underbodyhood	YARD	Guarantee value: Workstation	Section	Outside section	PLANT	Faults recorded: Section (1M)	Outside section (1M)	CUSTOMER (3M)	Quality guarantee level: WORKSTATION	Section	Outside section	PLANT	Comments
	Cargo drop	18.10.19	Cargo bolt not fit	v1 5				□ 1												⊙ 3								□ 1					1	4	1	5				nok	nok	nok	ok	

8. Standardization

8.3 Additional Audit / Check

6	Check point at wheel alignment Under Pit sheet.	10.10.19	Nilesh Deore	Closed

FAI UNDERPIT BUY-OFF - CHECK POINTS

Rev no:- 47 | Revision on - 10.10.19

Sr. No.	SC CC	Description	What to Check	V1 V2	How to Check	Source of complaint
1		Breather pipe checking	Fully insert No kink	V1	Marking Visual	RFI ANOVAC
2		Cargo regestration plate socket connection	Positive locking	V1	Pull-Push marking	RFI
3		Cargo structure on chassis	No loose bolt	V1	Marking Visual	
4		AT underbody wire	Routing on chassis	V1	Visual	RFI
5	CC	Diesel hose clip fitment checking	Fully tight and no cross fitment	V1	Visual	A Nova C
5	CC	Cargo bolt checking	Fully tight and no cross fitment	V	Visual	RFI
6		Fog lamp wire loom	Not fouling to exhaust, not hanging for non fog lamp vehicle	V1	Visual	A Nova C
7		Rear bumper fog lamp & RPAS socket connections	Positive locking, clipping on chassis	V1	Pull-Push	RFI
8		Clutch tube routing	Correct routing through clip Not fouling to Chassis(S3)	V1	Marking	RFI
9	SC	Front bumper push pin fitment	Not Loose As per model - S2 without frt. Splash guard	V2	Visual Relative Marking on Push pin	RFI
10	SC	Fuel gauge socket connection	Positive locking	V2	paint marking.	RFI

9. Summaries & Plan future actions

9.1 Conclusion:

Cargo Bolt not fitted is eliminated after

- Part corrected as per drawing – Slot provided at L2 and R2 cargo bolt fitment area.
- Outrigger position corrected in Z direction at chassis.

9.2 Horizontal Deployment:

NA

9.3 Lessons Learnt:

Things Gone Wrong :

Supplier not supplied the material as per updated drg..

Things Gone Right : .

Elimination of sealer application at mtg area has given good improvement.

Thank You.....